NLRI Error handling

draft-decraene-idr-nlri-error-handling-01

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BGP error handling as per RFC 7606

Error-handling approaches:

• Attribute discard	(if no impact on routing)
• Treat-as-withdraw	(if all NLRI can be read)
• Session reset	(otherwise / if one NLRI can't be parsed)

→ Parsing all NLRIs is critical to avoid session reset

NLRI with non-key data

Some AFI/SAFI encodes NLRI in MP_REACH_NLRI with both

- Key (“prefix”)
- Non-key data (kind of per NLRI attribute)

Examples of non-key data encoded in NLRI:

- Labeled Unicast: MPLS label stack
  - Including BGP CT, IP VPN
- EVPN: MPLS label(s), (Ethernet Segment Identifier)
- BGP CAR: TLVs

```
0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  NLRI Length  |  Key Length   |   NLRI Type   |              //
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+              //
|                  Type-specific Key Fields                    //
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                       Non-Key TLVs Fields                    //
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

MP_REACH_NLRI error handling with non-key data

Non-key data in NLRI increases:

- length of data to parse
- parsing complexity (multiple fields, TLVs, MPLS end of stack bit…)
- → chance of MP_REACH_NLRI parsing error
- → BGP session reset

BGP session restart is not likely to help

- Same routes, same NRLIs advertised
- → same error
- → BGP session reset again (and again and again)

→ High network impact, urgent human intervention required

- Likely urgent software fix, validation, deployment

Proposal (1)

- New non-transitive BGP attribute: “NLRI_KEY_LIST”
  - Same format as MP_UNREACH_NLRI (aka withdraw)
  - Encoded first in the UPDATE, followed by MP_REACH_NLRI

- If error condition in MP_REACH_NLRI preventing NLRI parsing
  - Perform “Treat-As-Withdraw”, using the NLRI_KEY_LIST attribute
  - Hence avoiding BGP session reset

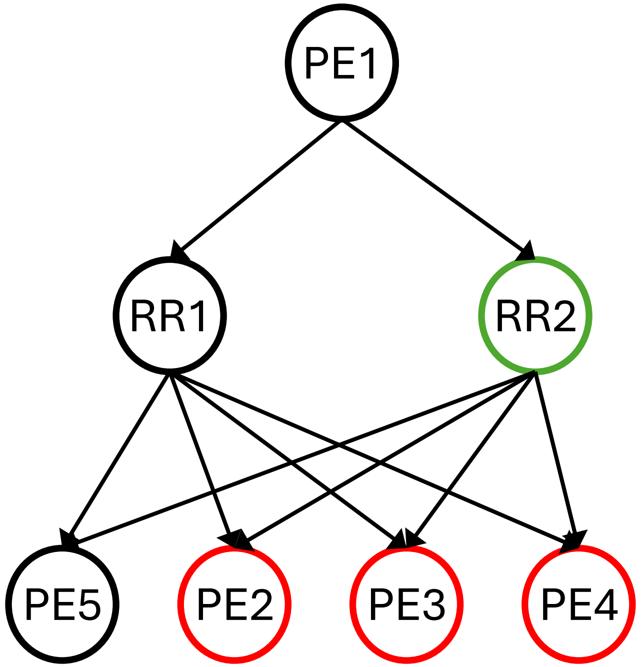
Proposal (2)

- Straightforward
  - (+) Reuse existing MP_UNREACH_NLRI format
    - (+) Sender constructs NLRI_KEY_LIST as a regular MP_UNREACH_NLRI
    - (+) Receiver reads NLRI_KEY_LIST as a regular withdraw/MP_UNREACH_NLRI
  - (-) Extra complexity on the sender (encodes twice the NLRI list)
    - Implementation, BGP UPDATE space, CPU time

- Optional.
- Cost is supported by the AFI adding the non-key data complexity

Use case example: BGP Label Unicast

- PE1 originates a BGP LU route
  - With a label stack of 2 MPLS labels
- MP_REACH_NLRI encoding
  - AFI: 1
  - SAFI: 4
  - Length of Next-Hop: 4
  - Next-Hop: 1.0.0.1
  - Reserved: 0
  - Length: 80 bits
  - Label1: 42
  - Bottom of stack1: 0
  - Label2: 52
  - Bottom of stack2: 1
  - Prefix: 1.0.0.1



A different color indicates a different implementation

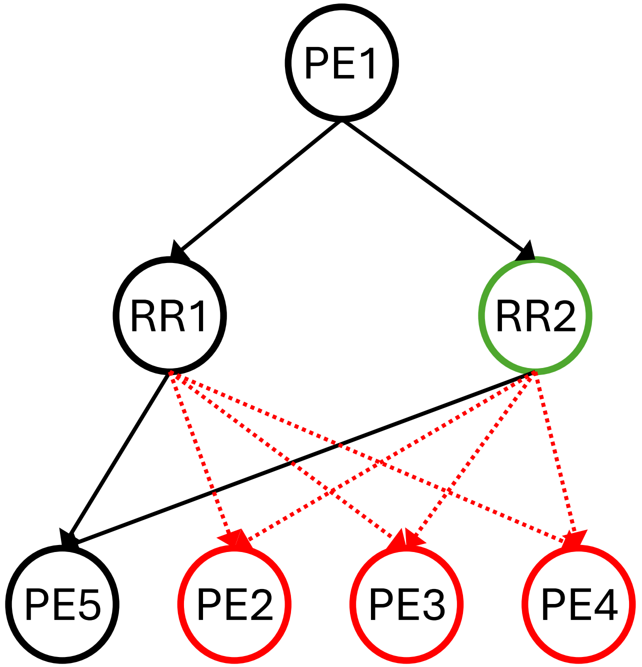
Current BGP error handling (w/ RFC 7606)

- Red implementation does not understand the label stack
  - Wrongly assumes a single label

- PE1 originates and RR1, RR2 propagate the route
  - Route is 100% correct as per RFC 3107

- PE2...PE4
  - can't parse the NLRI in MP_REACH_NLRI
  - can't locate IP prefix (key)
  - → can't treat-as-withdraw
  - → session reset

- RR redundancy does not help
  - RR code diversity does not help



A different color indicates a different implementation

New proposal

BGP UPDATE message

- NLRI_KEY_LIST
  - AFI: 1
  - SAFI: 4
  - Length: 56 bits
  - Compatibility field: 0x800000
  - Prefix: 1.0.0.1
- MP_REACH_NLRI
  - AFI: 1
  - SAFI: 4
  - Length of Next-Hop: 4
  - Next-Hop: 1.0.0.1
  - Reserved: 0
  - Length: 80 bits
  - Label1: 42
  - Bottom of stack1: 0
  - Label2: 52
  - Bottom of stack2: 1
  - Prefix: 1.0.0.1

PE2…PE4

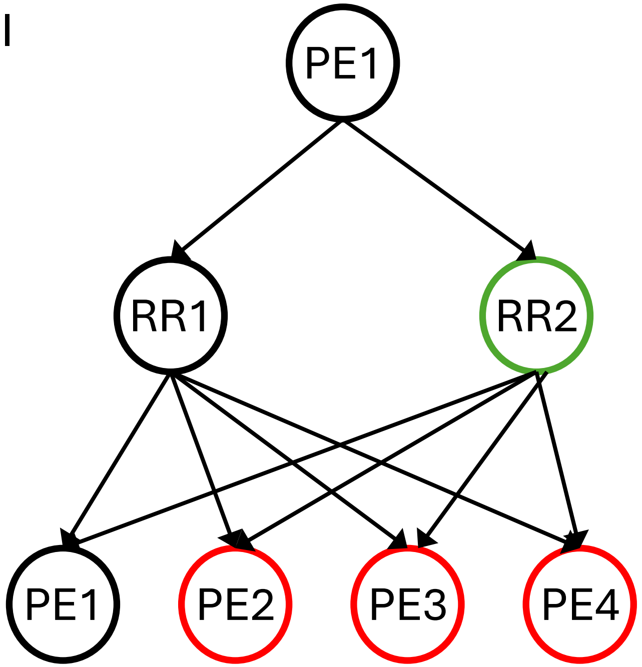
- can’t parse the NLRI in MP_REACH_NLRI
- can’t locate IP prefix (key)
- Error handling: reads NLRI_KEY_LIST as a MP_UNREACH_NLRI

→Effectively “Treat-as-withdraw”

→No session reset

Builds on MP_UNREACH_NLRI being simpler than MP_REACH_NLRI

- Shorter in size, Fixed size, less fields
- no non-key data



Next steps

- WG review and comments are welcomed
  - Thanks for reviews and comments on -00
  - -01 published

- Asking for WG adoption call